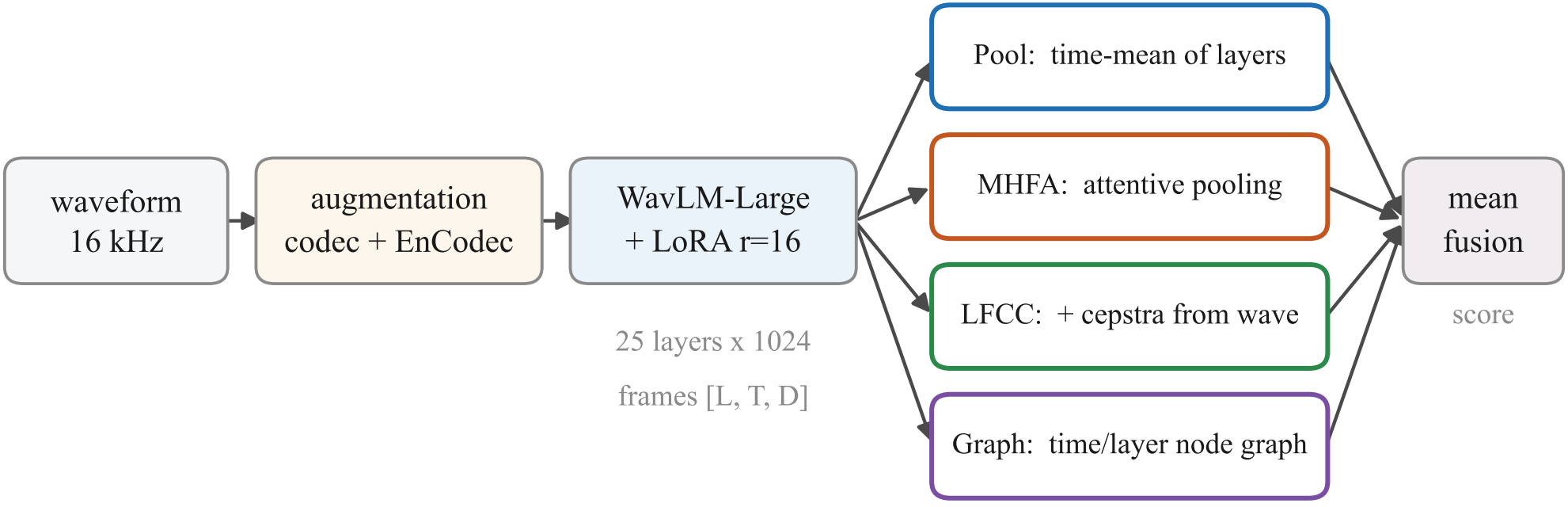four back-ends on ONE frozen-feature stack